

¹¹³Cd (Cd I), nuclear spin $I = 1/2$. Strongest classified transitions below 2 μm (NIST): 15 levels, 21 lines.

Energy level diagram for the 5s5p configuration of the neutral indium atom (In I). The diagram shows energy levels for various configurations and transitions between them.

Configurations and Energy Levels:

- 5s² 1S₀**: 0.000 cm⁻¹ (stable)
- 5s6s 3S₁**: 51483.980 cm⁻¹
- 5s5p 3P₂**: 304826.951 cm⁻¹ (τ = 1.47 s*)
- 5s5p 3P₁**: 30456.087 cm⁻¹ (τ = 2.69 μs)
- 5s5p 3P₀**: 30452.990 cm⁻¹
- 5s6d 3D₃**: 65367.227 cm⁻¹
- 5s6d 3D₂**: 65358.881 cm⁻¹
- 5s6d 3D₁**: 65353.372 cm⁻¹
- 5s5d 3D₃**: 59515.990 cm⁻¹
- 5s5d 3D₂**: 59497.868 cm⁻¹
- 5s5d 3D₁**: 59485.768 cm⁻¹
- 5s6d 1D₂**: 67838.401 cm⁻¹
- 5s5d 1D₂**: 59219.734 cm⁻¹
- 5s5p 1P₁**: 43692.389 cm⁻¹ (τ = 1.6 ns)

Transitions:

- 5s² 1S₀ → 5s6s 3S₁**: 4467.946 nm | 1.404 e_{ee}
- 5s² 1S₀ → 5s5p 3P₂**: 4480.323 nm | 2.592 e_{ee}
- 5s² 1S₀ → 5s5p 3P₁**: 4506.724 nm | 3.304 e_{ee}
- 5s² 1S₀ → 5s5p 3P₀**: 332.072 nm | 0.147 e_{ee}
- 5s² 1S₀ → 5s5p 1P₁**: 228.873 nm | 3.331 e_{ee}
- 5s5p 3P₂ → 5s6d 3D₃**: 283.173 nm | 0.973 e_{ee}
- 5s5p 3P₂ → 5s6d 3D₂**: 288.223 nm | 0.981 e_{ee}
- 5s5p 3P₂ → 5s6d 3D₁**: 298.120 nm | 2.324 e_{ee}
- 5s5p 3P₂ → 5s5d 3D₃**: 361.154 nm | 4.669 e_{ee}
- 5s5p 3P₂ → 5s5d 3D₂**: 361.548 nm | 0.562 e_{ee}
- 5s5p 3P₂ → 5s5d 3D₁**: 346.716 nm | 3.513 e_{ee}
- 5s5p 3P₁ → 5s6d 3D₃**: 346.895 nm | 2.035 e_{ee}
- 5s5p 3P₁ → 5s6d 3D₂**: 340.463 nm | 2.121 e_{ee}
- 5s5p 3P₁ → 5s6d 3D₁**: 352.000 nm | 2.009 e_{ee}
- 5s5p 3P₁ → 5s5d 3D₃**: 460.330 nm | 0.562 e_{ee}
- 5s5p 3P₁ → 5s5d 3D₂**: 460.330 nm | 0.562 e_{ee}
- 5s5p 3P₁ → 5s5d 3D₁**: 460.330 nm | 0.562 e_{ee}
- 5s5p 3P₀ → 5s6d 3D₃**: 332.072 nm | 0.147 e_{ee}
- 5s5p 3P₀ → 5s6d 3D₂**: 332.072 nm | 0.147 e_{ee}
- 5s5p 3P₀ → 5s6d 3D₁**: 332.072 nm | 0.147 e_{ee}
- 5s5p 3P₀ → 5s5d 3D₃**: 332.072 nm | 0.147 e_{ee}
- 5s5p 3P₀ → 5s5d 3D₂**: 332.072 nm | 0.147 e_{ee}
- 5s5p 3P₀ → 5s5d 3D₁**: 332.072 nm | 0.147 e_{ee}
- 5s5p 1P₁ → 5s6d 1D₂**: 414.147 nm | 0.309 e_{ee}
- 5s5p 1P₁ → 5s5d 1D₂**: 460.366 nm | 1.173 e_{ee}
- 5s5p 1P₁ → 5s5d 1D₂**: 644.025 nm | 6.236 e_{ee}

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J e J' \rangle $ (ea ₀)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$5s^2\ ^1S_0 - 5s5p\ ^3P_0^o$	228.8728	228.8023	1309.8647	3.331	6.250e+08	99.47	100	228.9 nm resonance line: first-stage MOT / slowing / imag...
$5s^2\ ^1S_0 - 5s5p\ ^3P_2^o$	314.1991	314.1081	954.1480	-	-	3.789e-09	-	314 nm ultra-narrow
$5s^2\ ^1S_0 - 5s5p\ ^3P_1^o$	326.1995	326.1055	919.0464	0.147	4.184e+05	0.06659	100	326.1 nm intercombination line: narrow-line MOT
$5s^2\ ^1S_0 - 5s5p\ ^3P_0^o$	332.0716	331.9760	902.7947	-	-	-	-	332 nm optical lattice clock transition
$5s5p\ ^3P_2^o - 5s5d\ ^3D_3$	361.1538	361.0508	830.0965	4.600	1.300e+08	-	-	361.2 nm metastable MOT
$5s5p\ ^3P_2^o - 5s5d\ ^3D_1$	361.5484	361.4453	829.1904	0.562*	4.510e+06	-	-	361.5 nm
$5s5p\ ^3P_1^o - 5s6s\ ^3S_1$	480.1254	479.9912	624.4045	2.592	4.100e+07	-	-	480.0 nm optical pumping / repumping

^{113}Cd hyperfine structure ($I = 1/2$)				
Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)	
5s5p $^3P_1^o$	-4313.860(13)	-	F=1/2: +4313.86	F=3/2: -2156.93
5s5p $^3P_2^o$	-3444.6344(16)	-	F=3/2: +5166.95	F=5/2: -3444.63
5s5p $^1P_1^o$	197.1(23)	-	F=1/2: -197.10	F=3/2: +98.55
5s6s 3S_1	-8108.6(9)	-	F=1/2: -8108.60	F=3/2: -4054.30
5s5d 3D_3	-2496.251(26)	-	F=5/2: +4992.50	F=7/2: -3744.38

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1^o$	228.873	106-114	1818(4)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1^o$	228.873	108-114	1336(3)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1^o$	228.873	110-114	865(3)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1^o$	228.873	112-114	408(3)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1^o$	228.873	116-114	-316(3)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1^o$	228.873	111-114	805(0.3)
$5s^2\ ^1S_0 - 5s5p\ ^1P_1^o$	228.873	113-114	344.9(5)
$5s^2\ ^1S_0 - 5s5p\ ^3P_2^o$	314.199	106-114	1916(4)
$5s^2\ ^1S_0 - 5s5p\ ^3P_2^o$	314.199	108-114	1404(3)
$5s^2\ ^1S_0 - 5s5p\ ^3P_2^o$	314.199	110-114	913.7(21)
$5s^2\ ^1S_0 - 5s5p\ ^3P_2^o$	314.199	112-114	429.7(16)
$5s^2\ ^1S_0 - 5s5p\ ^3P_2^o$	314.199	116-114	-320.2(17)
$5s^2\ ^1S_0 - 5s5p\ ^3P_2^o$	326.200	106-114	1913.0(10)
$5s^2\ ^1S_0 - 5s5p\ ^3P_2^o$	326.200	108-114	1402.4(10)
$5s^2\ ^1S_0 - 5s5p\ ^3P_1^o$	326.200	110-114	914.7(10)
$5s^2\ ^1S_0 - 5s5p\ ^3P_1^o$	326.200	112-114	429.9(10)
$5s^2\ ^1S_0 - 5s5p\ ^3P_1^o$	326.200	116-114	-321.5(10)
$5s^2\ ^1S_0 - 5s5p\ ^3P_1^o$	326.200	111-114	861.6(10)
$5s^2\ ^1S_0 - 5s5p\ ^3P_1^o$	326.200	113-114	370.8(10)
$5s^2\ ^1S_0 - 5s5p\ ^3P_0^o$	332.072	106-114	1911(5)
$5s^2\ ^1S_0 - 5s5p\ ^3P_0^o$	332.072	108-114	1404(4)
$5s^2\ ^1S_0 - 5s5p\ ^3P_0^o$	332.072	110-114	914.1(21)

$$\text{Shift} = v(\text{first isotope}) - v(\text{second isotope}).$$